The Regulatory System

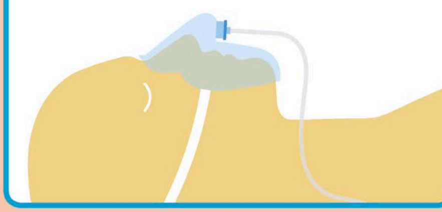
The human body is a cooperative of 38 trillion cells organized into different systems. Keeping them functioning at their best is a system of feedback mechanisms controlled by the brain.

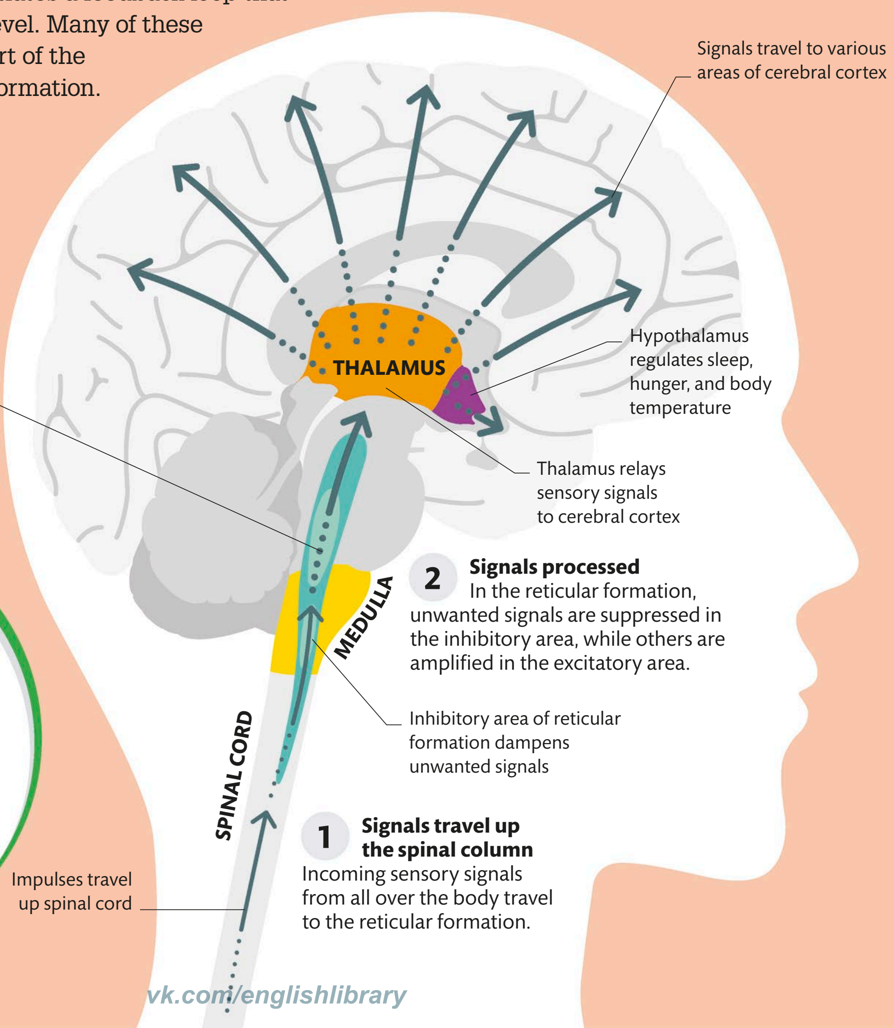
Maintaining stability

The process of maintaining a stable internal environment is called homeostasis. Key functions, such as breathing, heart rate, pH, temperature, and ion balances have to be kept within strict operating limits to prevent us from becoming ill. As the body works, its systems are constantly being moved away from their balance or set point (the value at which a system works best). When the change becomes too great, the body initiates a feedback loop that returns the system to its ideal level. Many of these functions are controlled by a part of the brain stem called the reticular formation.

GENERAL ANESTHETICS

A vital part of modern surgery, how general anesthetics work is not fully understood. What is known is that they act on the reticular activating system (comprising the reticular formation and its connections) to suppress awareness and on the hippocampus to temporarily suspend memory formation. Anesthetics also affect the nuclei of the thalamus, preventing the flow of sensory information from the body to the brain.





WHAT IS THE RETICULAR FORMATION?

The reticular formation consists of more than 100 nuclei that project to the forebrain, cerebellum, and brain stem, controlling many of the body's vital functions.